

3(a)	the point where (all) the weight (of the body) is taken to act	B1
3(b)(i)	vertical component = $54 \sin 35^\circ$ = 31 N	A1
3(b)(ii)	the (line of action of the) force (at B) passes through (point) A or the (line of action of the) force (at B) has zero (perpendicular) distance from (point) A	B1
3(b)(iii)	$54 \sin 35^\circ \times 0.68$ or $54 \cos 35^\circ \times 0.68$ or $W \times 0.34$	C1
	$54 \sin 35^\circ \times 0.68 + 54 \cos 35^\circ \times 0.68 = W \times 0.34$ so $W = 150$ (N)	A1
3(b)(iv)	total vertical force = $150 - 31$ = 120 N	A1
3(c)	$(\Delta)E = mg(\Delta)h$	C1
	$E = \frac{1}{2}mv^2$	C1
	ratio = $(m \times 9.81 \times 4.8) / (\frac{1}{2} \times m \times 9.2^2)$ or $(9.81 \times 4.8) / (\frac{1}{2} \times 9.2^2)$	C1
	= 1.1	A1